

■■■ Bems Farms — Chef Bems AI Chat & Add-to-Cart Flow Specification

Engineering Reference: Multi-Turn Conversational Lifecycle, Webhook Contract & Auto-Cart Injection

Client Trigger: Web Chat Widget & `/chef-bems` Page	Backend Bridge: Express API (`POST /api/ai/chef-chat`)
n8n AI Webhook: `https://n8n.srv1987482.hstgr.cloud/webhook/chef-bems`	Cart Mechanism: Client-Side React `CartContext` Auto-Injection

1. System Topology & Three-Tier Architecture

Chef Bems does not speak directly to the frontend. The Express backend acts as the secure, authenticated bridge in the middle to protect API keys, prevent browser CORS errors, record audits, and enable instant local AI failover.

1. Customer Frontend	2. Bems Farms Backend	3. n8n AI Automation Agent
• React Web & Mobile Chat UI • Attaches JWT Bearer Auth • Manages active `CartContext` • Displays real-time toast & badge	• Verifies customer identity (ID, email) • Forwards prompt + history to n8n • Saves conversation log & audits • Auto-fails over to Gemini if offline	• Processes prompt with LLM • Maintains contextual memory • Detects 'Add to Cart' intents • Returns structured `recipeBundle`

2. Multi-Turn Conversational Lifecycle & Add-to-Cart Stages

Customers engage in natural discussions (asking questions, modifying portions, discussing dietary health) across several rounds before explicitly deciding to add ingredients to their cart.

Stage & Turn	Customer Message & Discussion	n8n AI Behavior & Response	Cart State
Stage 1: Discovery (Round 1)	<i>'What healthy soup can I cook for a diabetic family member?'</i>	Suggests Okro or Afang Soup due to low glycemic index and high fiber. Gives general nutritional advice.	No Action Cart remains ■ 0
Stage 2: Refinement (Round 2)	<i>'How do I make the Okro Soup for 5 people? Can I use Catfish instead of beef?'</i>	Refines recipe for 5 persons. Explains steps for preparing fresh Catfish and Ugu leaves. Preserves history.	No Action Cart remains ■ 0
Stage 3: Cart Trigger (Round 3)	<i>'Awesome! Please add all those ingredients to my cart now.'</i>	AI detects Add to Cart Intent . Sets action: <code>'AUTO_ADD_TO_CART'</code> and returns 4 catalog items in JSON.	AUTO-INJECT Cart: ■ 0 ➔ ■ 4 (■13,500)

3. The 3 Add-To-Cart UI Execution Modes

Depending on the payload returned by n8n, the frontend handles cart operations in 3 versatile ways:

Option	Trigger Condition	Frontend Behavior & UI Feedback
Option 1: 1-Click Bundle Card	`recipeBundle` returned with `action: null` or omitted.	Renders an interactive card with checkboxes for each ingredient. User can uncheck items they already own and tap ■ Add Selected to Cart .
Option 2: Conversational Auto-Add	`action: 'AUTO_ADD_TO_CART'` in response.	Instantly adds all bundle items into active cart in the background. Shows animated toast: ■ Chef Bems added X items to your cart! .
Option 3: Instant Express Checkout	Customer taps ■ Cook Tonight on Recipe Card.	Loads bundle items into cart and immediately redirects browser to `/checkout` for 1-step direct payment.

4. n8n Webhook Input & Output JSON Contracts

4.1 Payload Sent by Bems Farms Backend to n8n Webhook

```
{
  "chatInput": "Awesome! Please add all those ingredients to my cart.",
  "message": "Awesome! Please add all those ingredients to my cart.",
  "userId": 42,
  "customerId": 42,
  "customerEmail": "ngozi@example.com",
  "sessionId": "user-42",
  "conversationHistory": [
    { "role": "user", "content": "What healthy soup can I cook for someone with high blood sugar?" },
    { "role": "assistant", "content": "I recommend Okro Soup! It is low in carbs and high in fiber." }
  ],
  "cartItems": [
    { "name": "Bems Brown Beans", "quantity": 1, "unit_price": 4500 }
  ]
}
```

4.2 Required JSON Output from n8n Workflow (To Trigger Cart Addition)

```
{
  "reply": "I have added all 4 fresh ingredients for Okro Soup directly into your shopping cart!",
  "action": "AUTO_ADD_TO_CART",
  "recipeBundle": {
    "recipe_name": "Nutritious Okro Soup (Serves 5)",
    "servings": 5,
    "items": [
      { "name": "Fresh Okro Fingers", "quantity": 1, "unit": "1kg basket", "price": 2500 },
      { "name": "Fresh Catfish (Cleaned & Cut)", "quantity": 2, "unit": "1kg pack", "price": 6200 },
      { "name": "Pure Red Palm Oil", "quantity": 1, "unit": "1 Litre bottle", "price": 2800 },
      { "name": "Fresh Ugu Leaves (Pumpkin)", "quantity": 2, "unit": "1 bunch", "price": 2000 }
    ]
  }
}
```

5. AI Automation Engineering Checklist

- **1. Webhook URL:** Configure webhook path on Hostinger Cloud to match <https://n8n.srv1987482.hstgr.cloud/webhook/chef-bems>.
- **2. Production Activation:** Ensure the workflow is toggled to '**Active**' in the top right corner of the n8n canvas.
- **3. Response Schema:** The final node in n8n must return JSON containing reply, action, and recipeBundle.
- **4. Intent Keyword Triggers:** Set action: 'AUTO_ADD_TO_CART' when user prompt contains: 'add to cart', 'buy these', 'put in cart', 'order ingredients'.
- **5. Product Naming:** Match item names to Bems Farms catalog (e.g., 'Ofada Rice', 'Palm Oil', 'Brown Beans', 'Garri', 'Fresh Tomatoes').